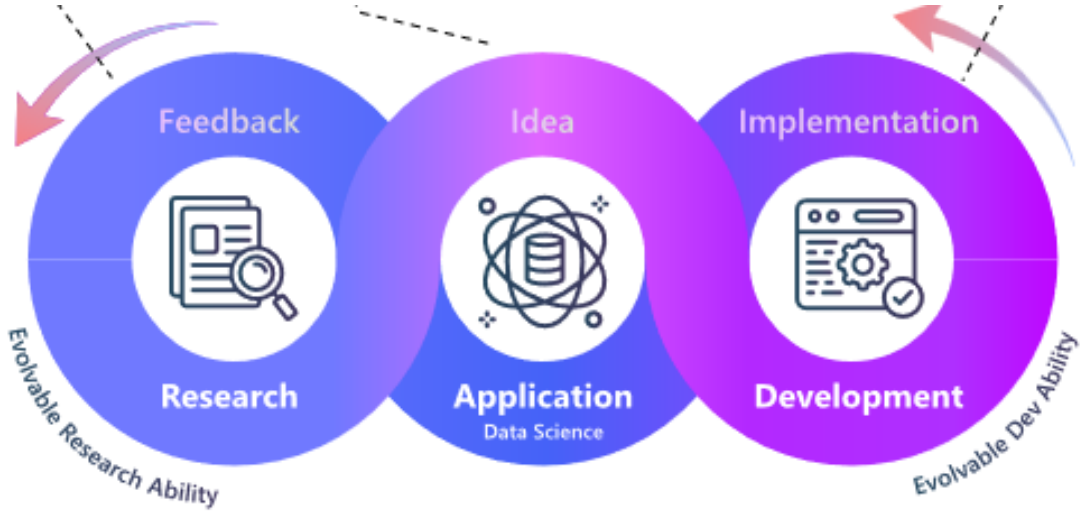


Agentic Research: 智能体时代的研究

——新范式与新挑战

For Group Meeting

3.13

Everyone is all-in

共识：Agent 是:

- **最有希望成功的AI商业模式之一**
- **最有希望达到AGI的范式之一**

What is Agent

Agents are systems that **independently** accomplish tasks on your behalf.

核心：通过 LLM + 工具调用，使得系统自动完成复杂功能和任务，使得AI 能够“**自主行动**”

Model

Tools

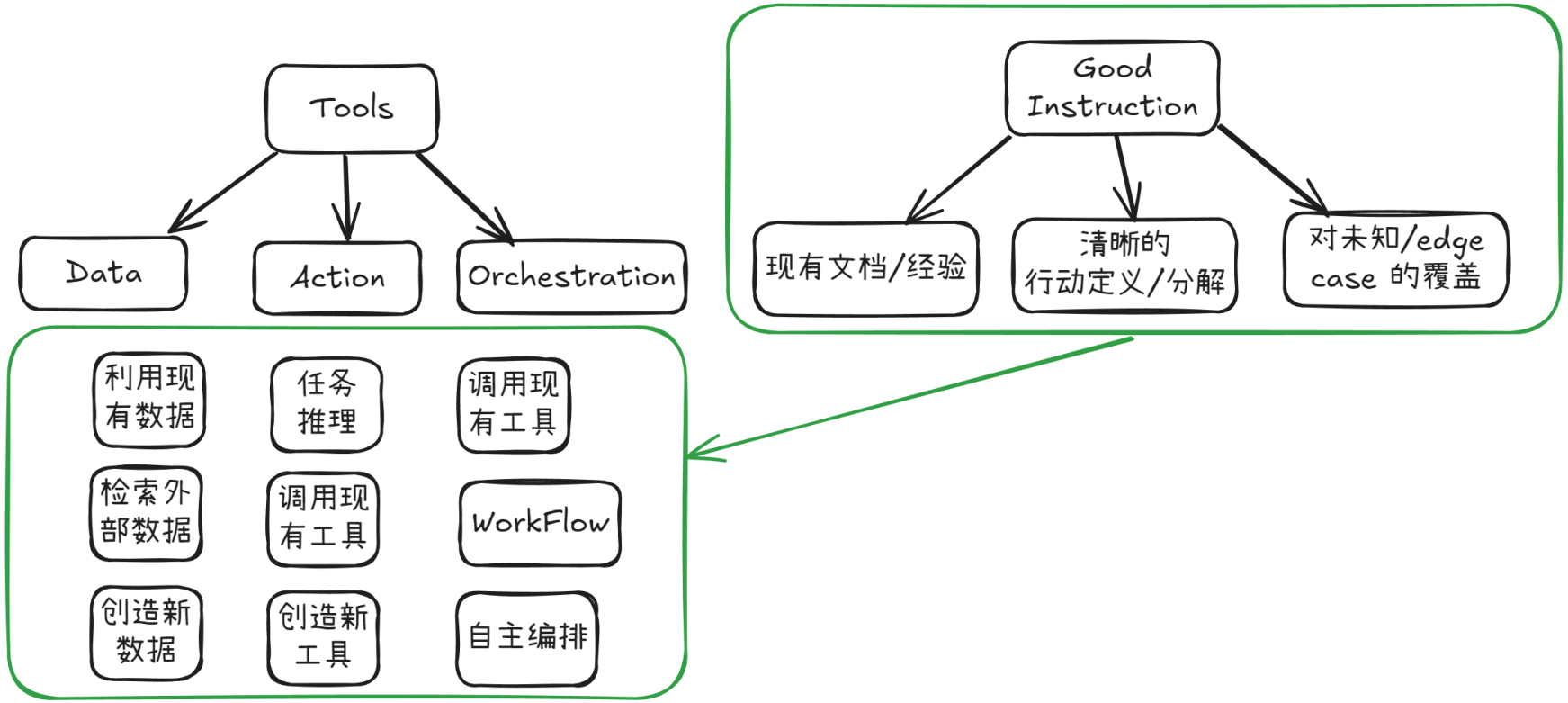
Instruction

Python

```python
weather_agent = Agent(
    name="Weather agent",
    instructions="You are a helpful agent who can talk to users about the weather.",
    tools=[get_weather],
)
```

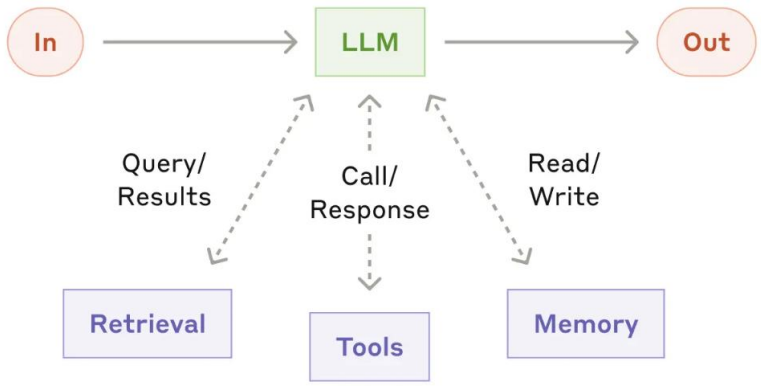
What is Agent

- 广义的“工具” （**Tools**） 是 Agent 的核心
- 清晰明确的“指令” 是 Agent 的基础



What is Agent

- 最简单的Agent block：LLM 增强

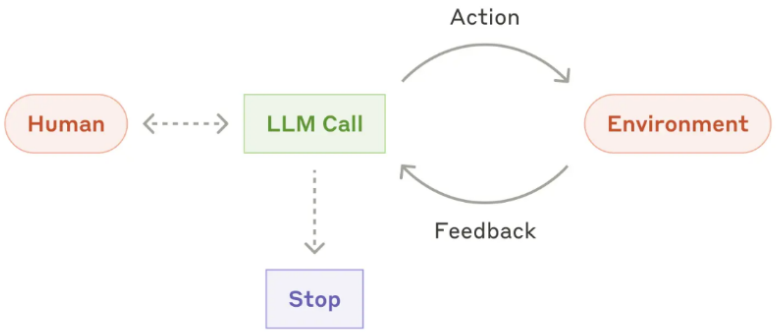


The augmented LLM

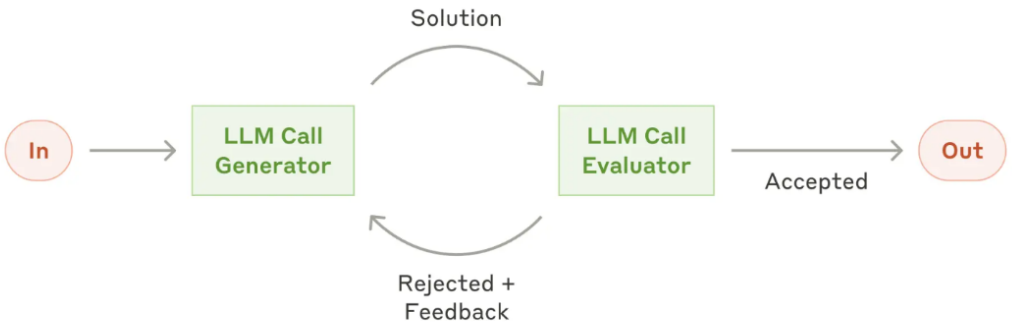
What is Agent

- **Orchestration （组织/编排）： 如何让多个Agent 协同合作**

分工型 orchestrator-workers

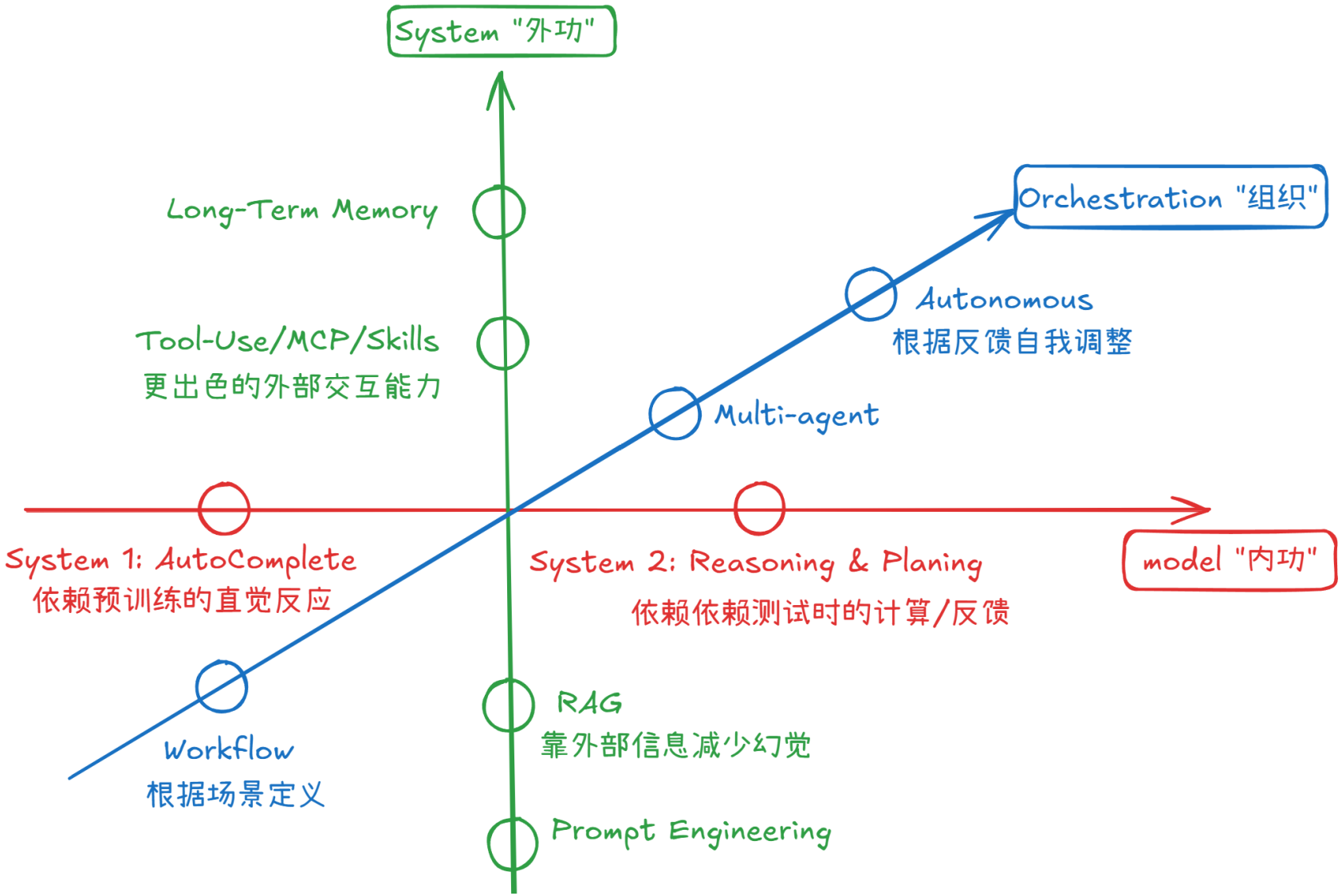


完全自主型 Autonomous agent

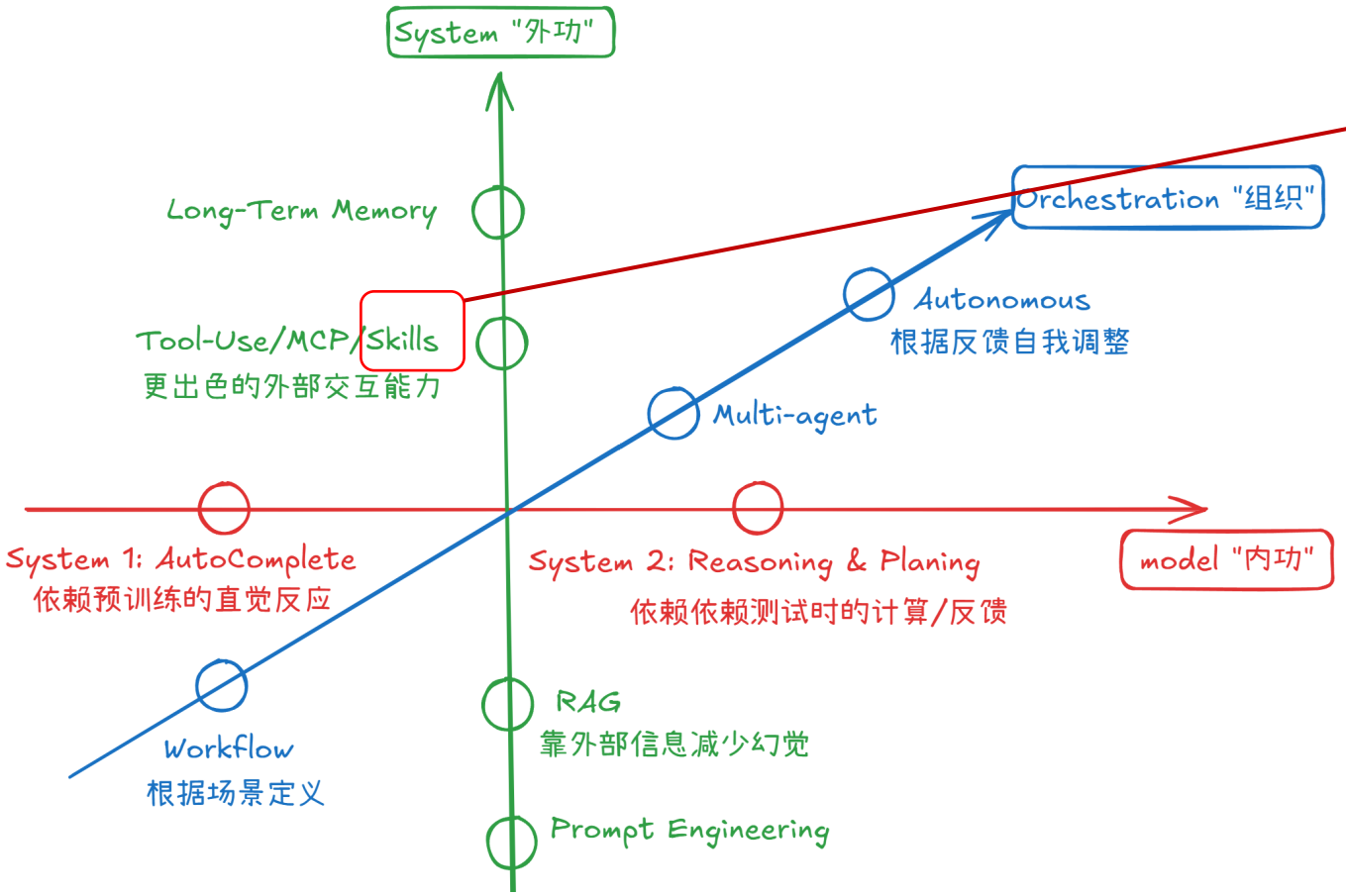


优化型 evaluator-optimizer workflow

当前Agent 技术象限



当前Agent 技术象限



- **“渐进式提示词披露”**

```
---
name: my-skill-name
description: A clear description of what this skill does and when to use it
---

# My Skill Name

[Add your instructions here that Claude will follow when this skill is active]

## Examples
- Example usage 1
- Example usage 2

## Guidelines
- Guideline 1
- Guideline 2
```

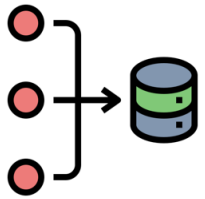
HIGH SCORES — 47 RECORDS

RANK	SKILL	AUTHOR	CATEGORY	JUDGES	SCORE
#4	breaking-change-detector	marcusgoll	DEVELOPMENT	AI	9
#5	backend-models-standards	maxritter	BACKEND	AI	9
#6	plotly	K-Dense-AI	DATA	AI	8.9
#7	data-transform	Starlitnightly	DATA	AI	8.9
#8	pytorch-lightning	K-Dense-AI	AI-ML	AI	8.9
#9	responsive-images	jezweb	DEVELOPMENT	AI	8.9
#10	verification-before-completion	obra	DEVELOPMENT	AI	8.9
#11	statsmodels	Microck	AI-ML	AI	8.9
#12	shopify-app-dev	Microck	DEVELOPMENT	AI	8.9
#13	design-tokens	marcusgoll	DEVELOPMENT	AI	8.9
#14	marketing-social-media	vasilyu1983	DEVELOPMENT	AI	8.9
#15	react-ui-patterns	ChrisWiles	FRONTEND	AI	8.9
#16	test-driven-development	brobertsaz	TESTING	AI	8.9
#17	geopandas	Microck	DATA	AI	8.9

SkillHub - Claude Skills & Agent Skills Marketplace | Claude Code Skills

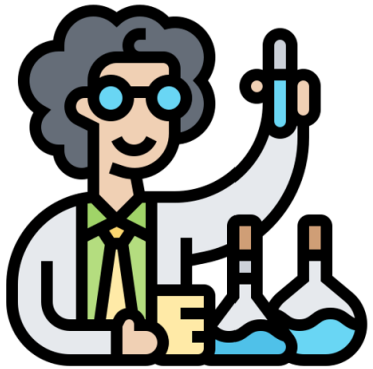
真实世界的挑战

- 真实世界的高价值任务 (如 企业界R&D, 科研)，**长链路，无清晰单一反馈，不确定性大**
- 此类任务的挑战：解一个高维、动态、不定的 **反向优化问题 （Inverse Opt. Problem)**

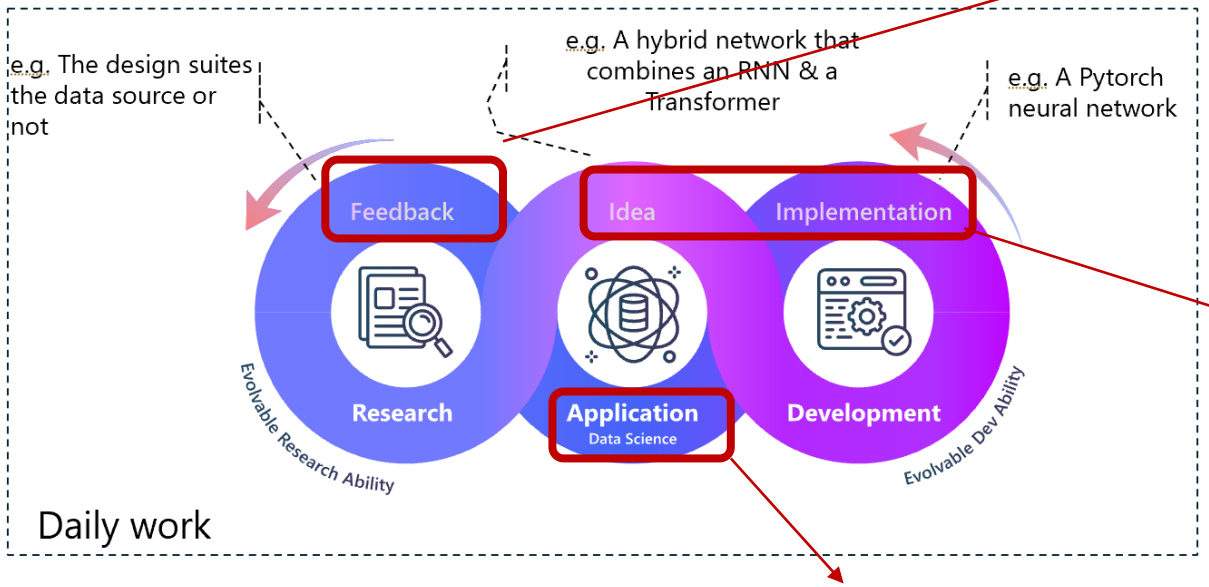


Unique data source

Objectively measurable task



Applied data scientist



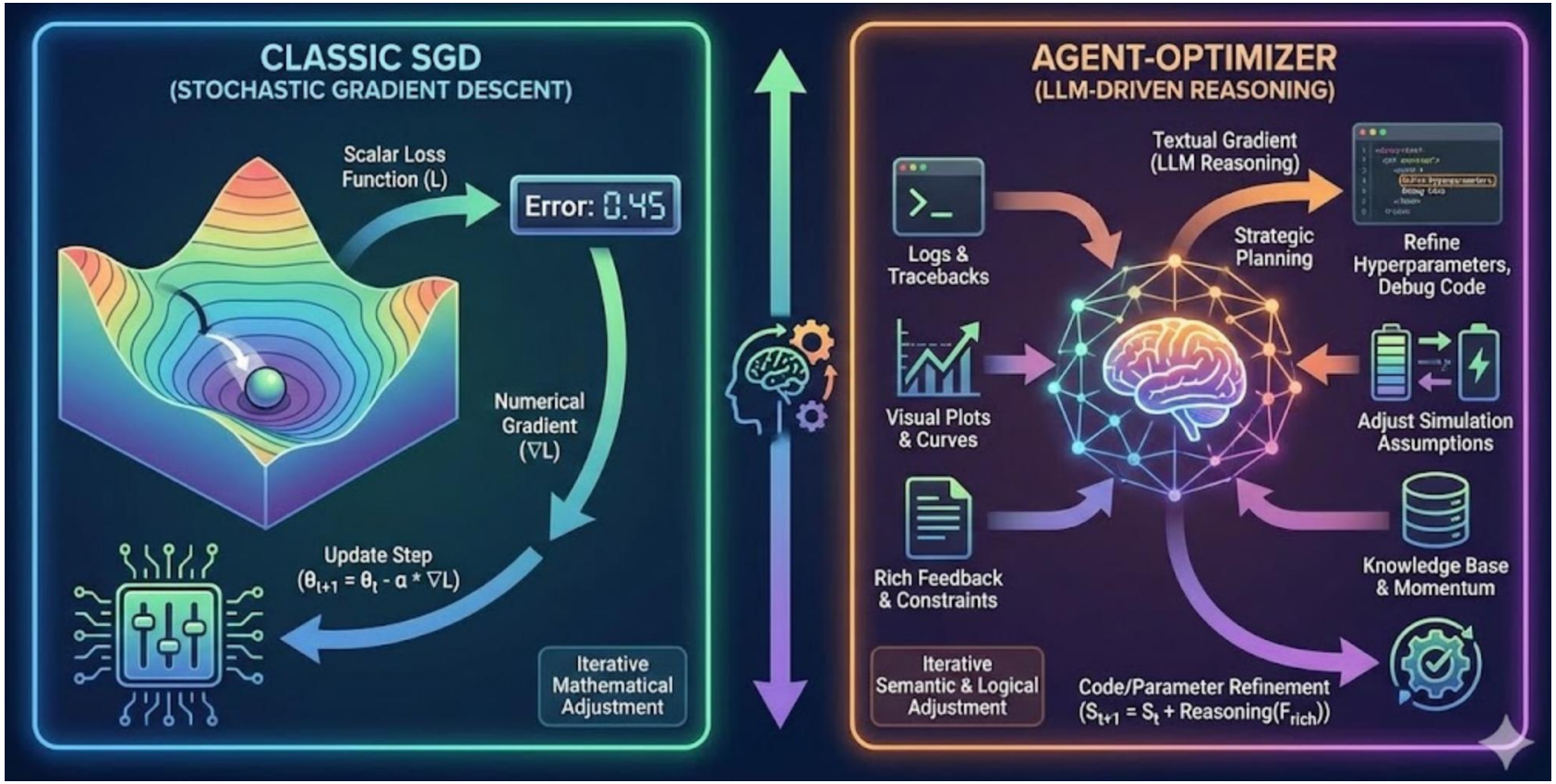
Rich Feedback
（不仅是个loss）

模型效果不好:
- idea 有问题?
- 实现有问题?
- 数据有问题?
- …

长周期、高成本、黑箱…

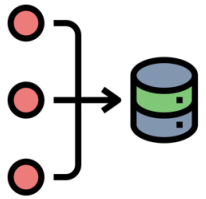
新范式：Agent as "Super Optimizer"

- 类比 SGD （梯度下降）算法，该如何定义&设计 Agentic Optimizer 中的损失、梯度、动量、更新?
- **Reasoning as Optimization**



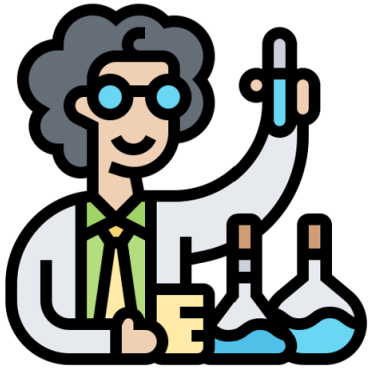
Project 1: R&D-Agent

- 为 **数据科学R&D** 自动化 设计
- 常见场景：给定 数据集 + 资源约束 + 简单任务说明，如何快速迭代最优方案?
- **Solution： Research Agent + Development Agent 协同进化**



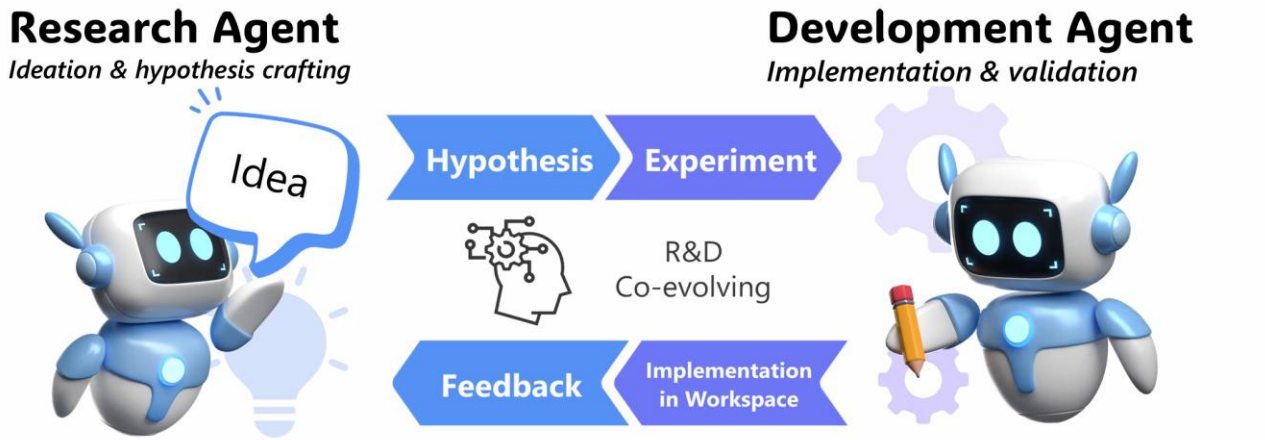
Unique data source

Objectively measurable task



Applied data scientist

Traditional Applied-Data-Science R&D: A Trial-and-Error Evolving
***R&D-Agent* codifies human experts' R-D Co-Evolution Loops**



Generates high-quality ideas & experiment plans

Produces robust, production-ready code

Project 1: R&D-Agent

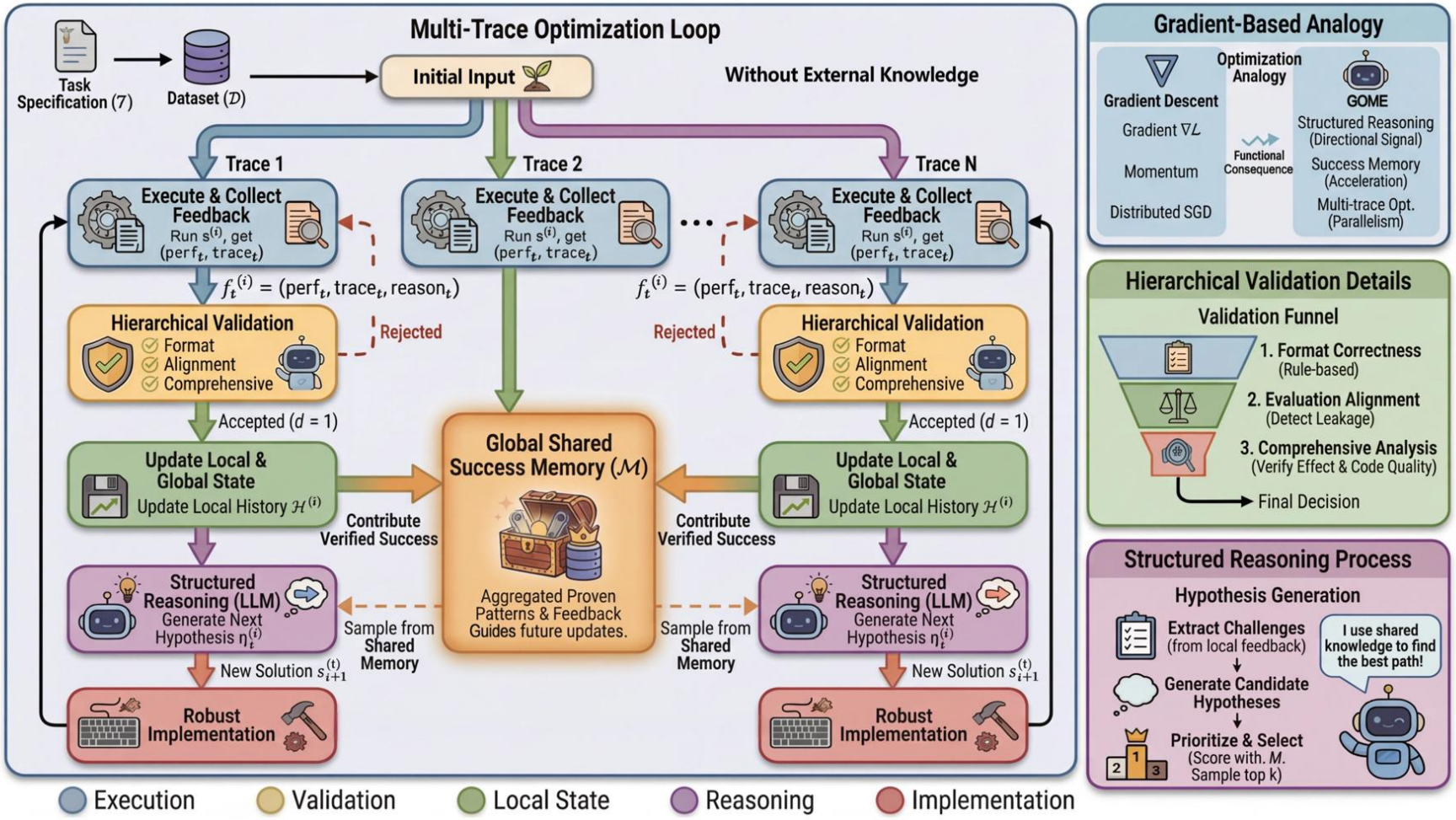
核心设计：

- 多路并行探索

- 全局共享记忆

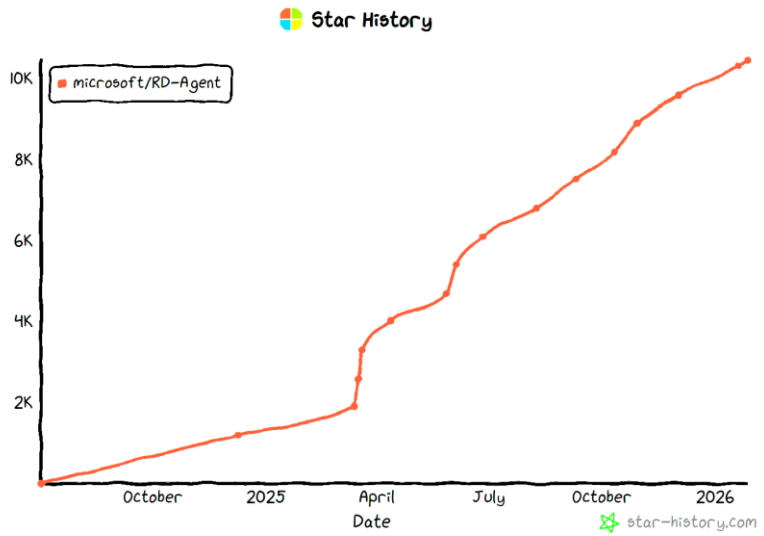
- 结构化反馈 + 推理优化

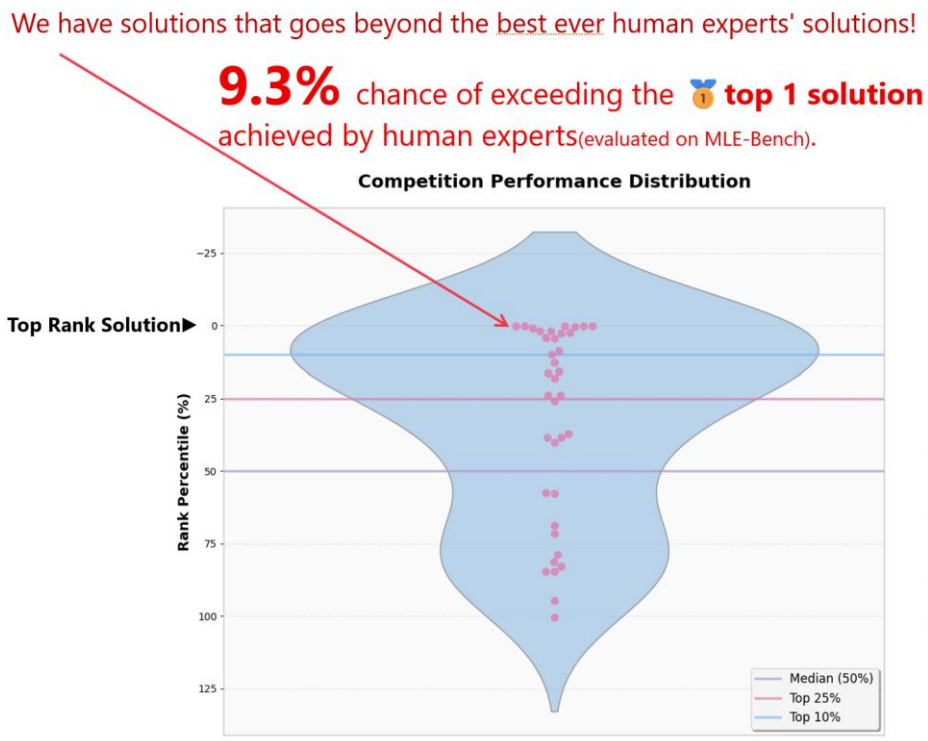
- 基于不确定性评估的假设&决策生成

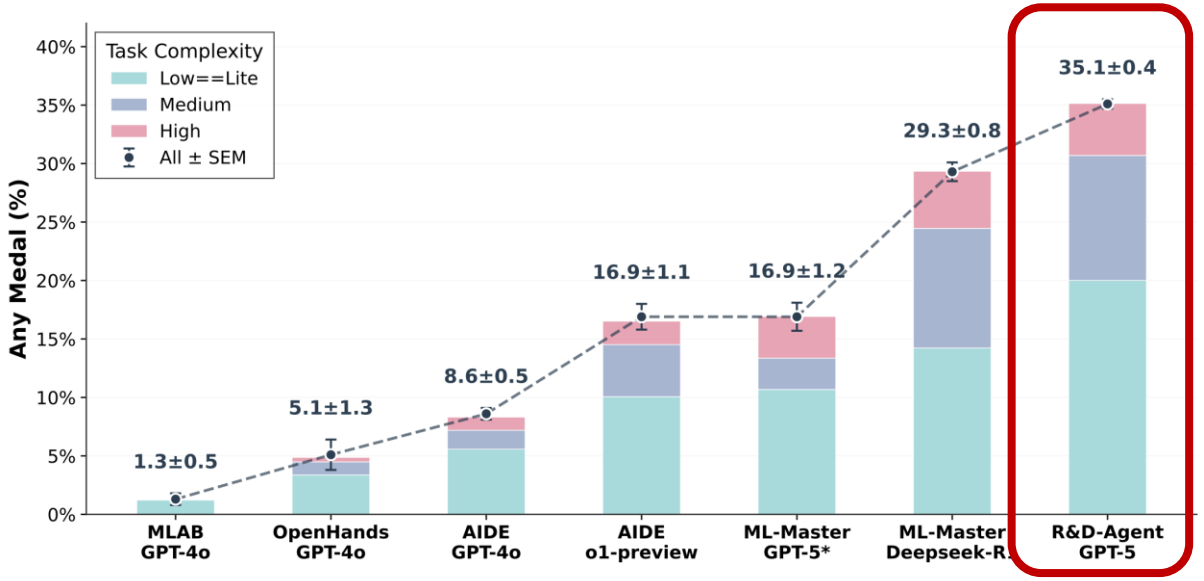


Project 1: R&D-Agent

- 全自动打 75+ Kaggle 比赛(包含 CV, NLP, 语音，时序等多样任务)
- 35% 拿牌率， 在OpenAI MLE-Bench 上位列所有开源模型 的第一名
- Arxiv: 2505.14738， Github: 10.3k+ star



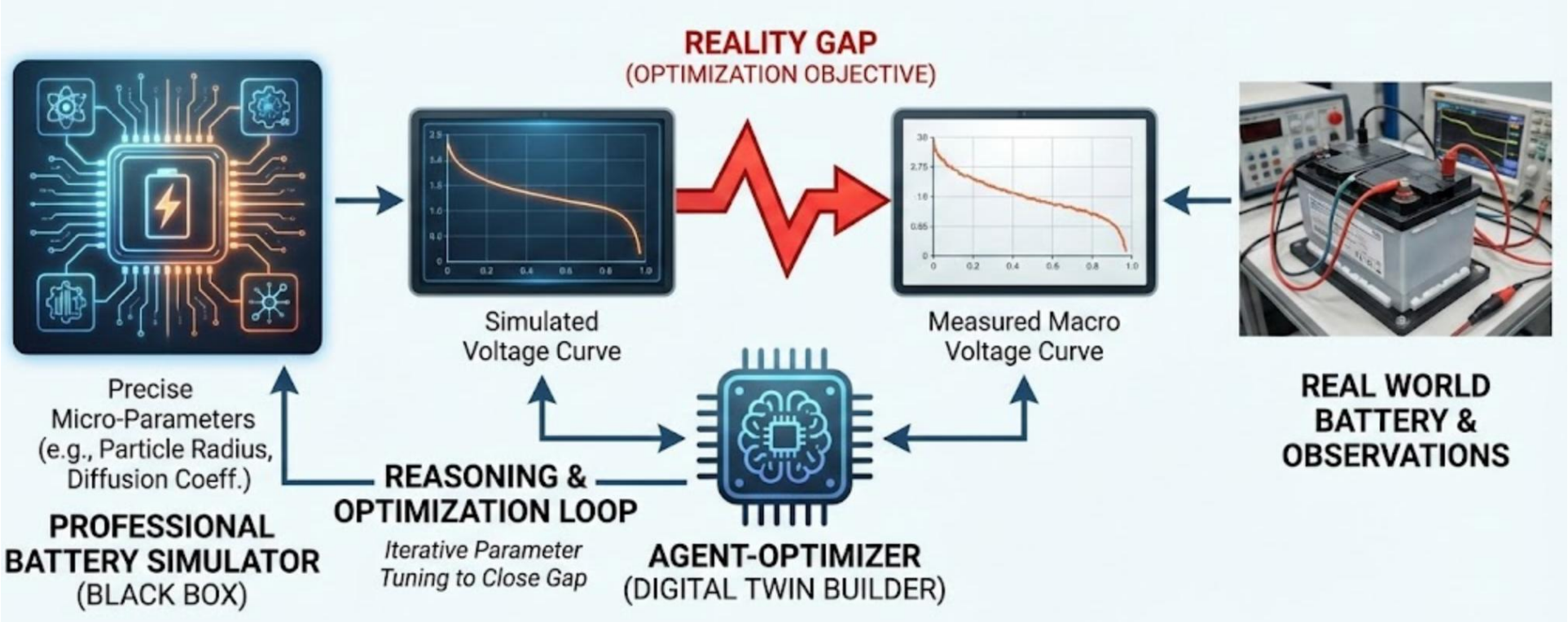




Project 2: Battery-Sim-Agent

以 **“电池研发”** 为代表的制造业 R&D 常见挑战:

- 从物理世界中获取反馈的 **时间长，成本高 （跑几千小时才能获得完整曲线）**
- 行业里有不错的 **“Digital Twin/仿真器（Simulator）”，但仿真器的设定 与 真实 存在GAP (Sim2Real Gap)**



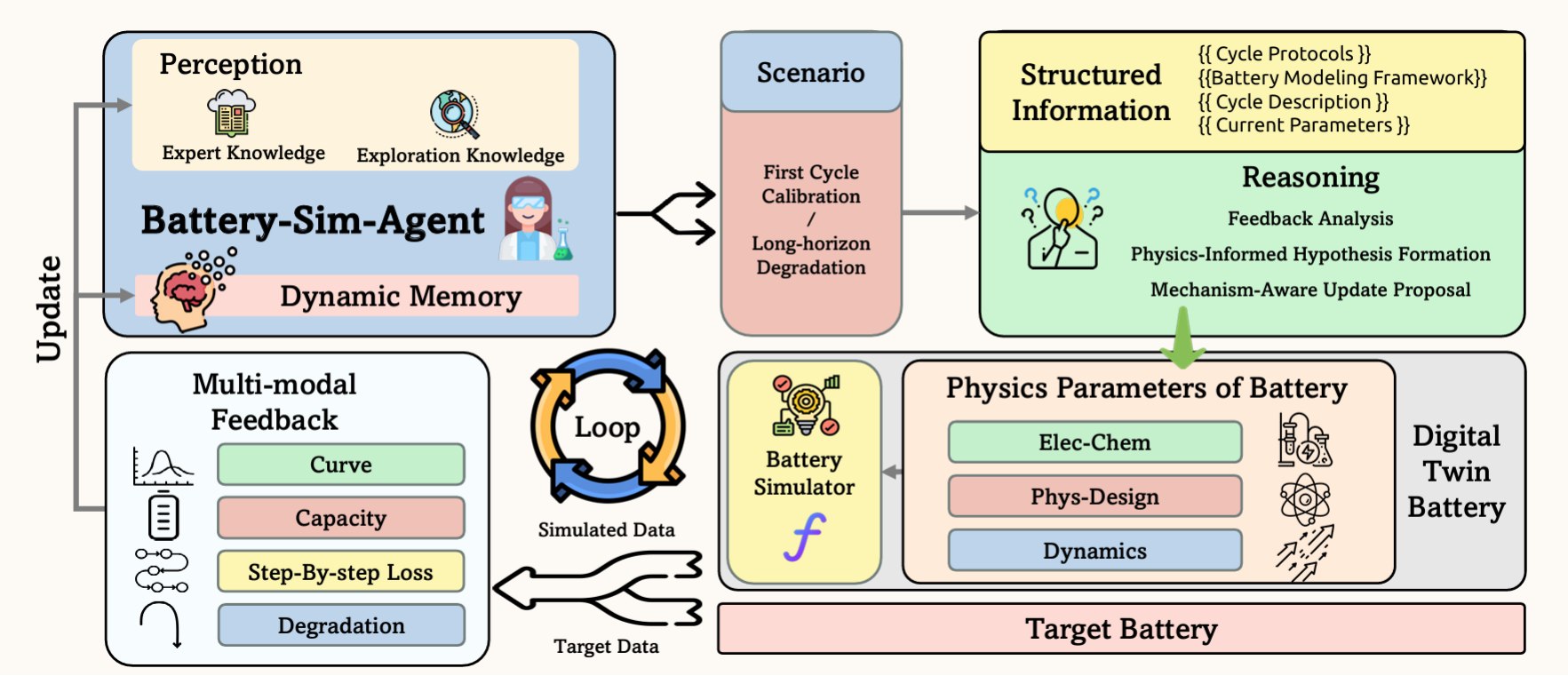
Project 2: Battery-Sim-Agent

核心设计：

- 把 simulator 和 真实部分观测的gap 作为 feedback

- 专业知识驱动先验

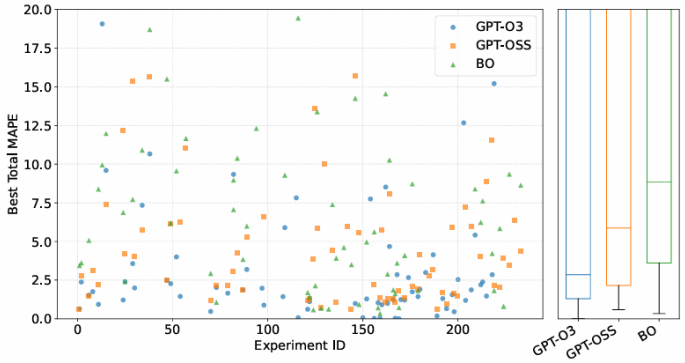
- 基于多模态反馈的推理



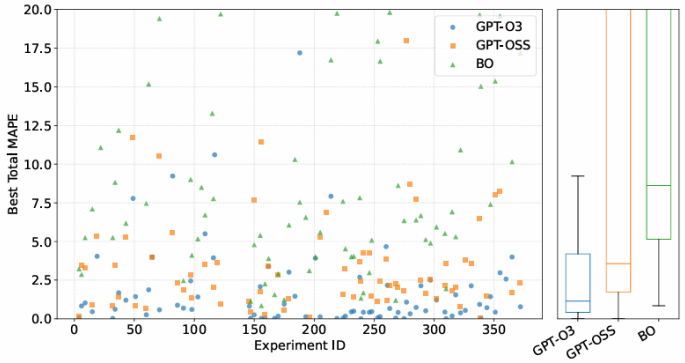
Project 2: Battery-Sim-Agent

- 比传统黑盒优化算法（贝叶斯优化、遗传算法等）的效果提升 5x+
- 在不常见的极端模式测试下更加稳定和鲁棒

Mode	Methods	Chen2020	ORegan2022	Ecker2015	Prada2013	Marquis2019
			MAPE			
Regular	Default	159.09±118.6	160.47±119.0	108.42±65.1	79.16±56.1	122.60±79.2
	BO	211.97±404.4	81.73±224.0	27.37±80.3	56.31±222.2	13.54±9.3
	BatterySimAgent-OSS	38.05±100.0	46.34±53.3	7.63±17.3	23.74±44.7	9.55±20.3
	BatterySimAgent-O3	**12.60±24.5**	**34.18±48.2**	**0.77±1.2**	**5.97±12.2**	**1.27±1.1**
Extreme	Default	119.32±110.4	181.95±181.2	100.43±115.3	150.65±87.9	108.00±89.7
	BO	159.41±362.4	84.55±213.7	137.31±278.6	**17.05±25.3**	**8.42±6.3**
	BatterySimAgent-OSS	23.66±42.2	50.88±61.5	45.47±92.2	23.96±42.3	19.50±39.1
	BatterySimAgent-O3	**23.38±52.4**	**19.44±24.2**	**27.85±79.5**	59.14±62.3	48.34±90.4



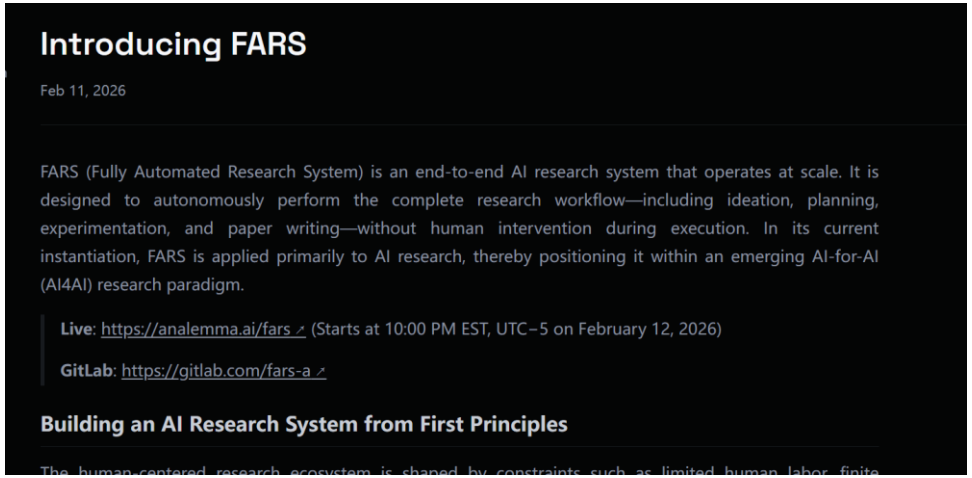
(a) Regular Mode



(b) Extreme Mode

Why everyone should learn & use Agent

- Agentic AI加持研究的自动化, 已经是不可逆的趋势



Introducing FARS

Feb 11, 2026

FARS (Fully Automated Research System) is an end-to-end AI research system that operates at scale. It is designed to autonomously perform the complete research workflow—including ideation, planning, experimentation, and paper writing—without human intervention during execution. In its current instantiation, FARS is applied primarily to AI research, thereby positioning it within an emerging AI-for-AI (AI4AI) research paradigm.

> **Live**: https://analemma.ai/fars (Starts at 10:00 PM EST, UTC−5 on February 12, 2026)
>
> **GitLab**: https://gitlab.com/fars-a

Building an AI Research System from First Principles

FARS - Analemma | Analemma

Why everyone should learn & use Agent

- 3-5 年之后的工业界/学术界，会和现在完全不一样
- 传统意义的 “研究者/硕博” 的很多 “价值” 会彻底消亡

对个人的影响

- 极大减少脏活累活 （codebase, data, baseline）
- 高效验证 POC (Proof of Concept)
- 高效交叉，一鱼多吃：
  - 一个核心 idea + 多场景应用和故事
- 快速进入一个领域的门槛降低到零

个体效率的极高提升

对学术系统的影响

- Paper 数量的爆炸， 同行评审制度的消亡
- Paper 不再是终点，影响力，System 才是
- 学科的边界逐渐模糊
- "一人成军"

3-5年内，整个评价系统的洗牌

Why everyone should learn & use Agent

"专家级别的深刻"
insights / Vision

- 对现有范式缺陷的洞察
- 对未来范式的预判

场景/需求/问题

- 金矿在看不见的地方
- "科幻场景驱动"

Agent Infra

- 抓住时间差和信息差，做第一批吃螃蟹的人
- 成为最会使用这些工具、甚至能制造更先进工具的人

How: 具体怎么做?

个人科研进度的结构化管理

- Code, Method, Design -> All In One
- Data, Task, Doc -> 接口标准化

团队组织的结构化管理

- 基于飞书（暂定）的交流 + Open-Claw-like-tool
- 组会讨论/头脑风暴的成果过程 结构化存储

研究之外流程的自动化尝试

- PR/PPT/Benchmark - 如何自动化
- 本子/基金/论文/审稿/画图 -> 尽量自动化

How: 具体怎么做?

个人可以立马做的事

- 把 Claude-Code/CodeX 玩起来（API 实验室报销）

- 尽量把手头的科研进度，结构化成 .md 文档 （所解决的问题，方法，进展）

- 代码结构和版本管理：尽可能地用规范的格式(.yaml as config + terminal line + notebook)和工具（Github）

最后预期形成的团队Infra

- Skills Set

- 能快速做概念验证的 Dataset hub, Baseline hub, Simulator hub

- 适合新同学快速上手的Tutorial on project/writing

- 论文润色，代码管理， PR。。 等一系列科研流程的 Agentic SOP

- "Idea 库"

Resources:

- a-practical-guide-to-building-agents.pdf
- mastering-ai-agents-galileo.pdf
- 构建高效的人工智能代理 \ Anthropic